Higher National Diploma in Engineering (Electrical)
04th Year, First Semester Examination – 2023
EE 3214 – Construction and Protection of Electrical systems

Instructions for Candidates:
Answer Any **Five (05)** questions.
Each question carries equal marks

No. of questions:06
No. of pages : 03
Time : **03 Hours**

Q1.

a) What is the Per Unit (PU) system? Explain its importance in electrical power systems.?

(05 Marks)

b) Discuss how the per unit system simplifies fault analysis in power systems.? A power system operates with multiple voltage levels (generation, transmission, and distribution). Explain how the per unit system is applied across these levels.

(08 Marks)

c) A three-phase transmission line has the following parameters:
Resistance: 10 Ω
Reactance: 40 Ω
Base Power: 100 MVA
Base Voltage: 220 kV

Calculate:
i. Base current
ii. Base impedance
iii. Per unit resistance and reactance

(07 Marks)

[Total 20 Marks]

a) .

I. Explain the significance of fault analysis in power systems?

II. List and briefly describe the types of faults that commonly occur in three-phase systems

(05marks)

b) Why is it essential to calculate fault current in power systems? Describe the role of protective relays and circuit breakers in reduce the effects of faults in power systems?

(06marks)

c) A 3-phase, 20 MVA, 10 kV alternator has an internal reactance of 5% and negligible resistance. Find the external reactance per phase to be connected in series with the alternator so that the steady current on short-circuit does not exceed 8 times the full load current (09 Marks)

[Total 20 Marks]

a) Discuss the common faults in transformers and List at least three causes of overheating in transformers?

(06 Marks)

b) What is Buchholz relay protection, and explain its working principle with suitable diagram?

(08 Marks)

c) A transformer with a primary voltage of 11 kV and a secondary voltage of 415 V is protected by a differential relay. Calculate the relay's operating current if the CT ratio is 200:1 on the primary and 500:1 on the secondary.

(06 Marks)

[Total 20 Marks]

II. Differential relays
III. Plug Setting Multiplier (P.S.M)
IV. Time Setting Multiplier (T.S.M) (08 ma

c) A 500 MW generator experiences a sudden unbalance in the phase curre during operation.

i. What protection mechanisms would detect this fault?
ii. How should the system respond to prevent equipment damage?

(06 m

[Total 20 Ma

Q5.

a) Explain the role of protection relays in switchgear systems? Describe th functioning of overcurrent and differential relays in protecting against sh circuits and overloads.

(06 M

b) With the help of neat diagram, Explain the construction and working of: Non-directional induction type overcurrent relay

(06 M

c) A 3-phase, 4-wire system supplies loads which are unequally distributed three phases. An analysis of the current flowing in R, Y, and B lines s that in the R line, the positive phase sequence component is $200\angle 0^\circ$ an negative phase sequence component is $100\angle 60^\circ$ A. The total observed cu flowing back to the supply in the neutral conductor is $300\angle 300^\circ$ A. Cal the currents in the three lines(R, Y and B).

(08 M

Q6. Write short notes on, [Total 20 Ma

a) Switchgear
b) Air Circuit Breaker
c) High Voltage Fuses
d) Reclosers
e) Lightning Protection Systems

(Marks 04

[Total 20 Mar